

AMAM — Assumption Materiality Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Assumption Standard (PAS)

Operational Layer: Prediction Assumption Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Assumption Materiality

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Assumption Materiality

Canonical Definition

Assumption Materiality Module (AMAM) defines the governance conditions under which Predictive Assumptions are assessed and classified according to the significance that their state, change, weakening, breach, uncertainty, or invalidity may have for the meaning, methodology applicability, probability, uncertainty, feasibility, interpretation, reliance conditions, or continuing governability of a prediction.

AMAM determines which assumptions are sufficiently consequential to require elevated governance attention.

It transforms:



Identified Assumptions + Dependency Structure + Sensitivity Profile

into:

Governed Assumption Materiality State

and distinguishes ordinary assumptions from assumptions whose condition is materially important to the prediction.

Its foundational distinction is:

Assumption Presence $\neq$ Assumption Materiality

Operational Role

AMAM serves as the fourth internal governance space of the Prediction Assumption Standard (PAS) .

It receives:

Identified Predictive Assumptions

Governed Assumption Dependency Map

Governed Assumption Sensitivity Profile

and determines the significance of each assumption within the governed prediction.

The module asks:

Which assumptions matter enough that their weakening, failure, uncertainty, or change could materially alter the legitimate meaning or use of the prediction?

The governed progression is:

Assumption Identity → Dependency Context → Sensitivity Context → Consequence Assessment → Materiality Determination → Criticality Determination → Governed Assumption Materiality Profile

The resulting profile becomes a principal input into:

ACGM — Assumption Change Governance Module.

Internal Module Flow

Predictive Assumption Identification → Assumption Dependency Mapping → Assumption Sensitivity → → Assumption Materiality → Assumption Change Governance

Current Position: Assumption Materiality

Module Operational Space

AMAM governs:

materiality assessment; materiality criteria; materiality dimensions; materiality state; predictive consequence; methodological consequence; probability consequence; uncertainty consequence; feasibility consequence; interpretation consequence; reliance consequence; decision consequence; temporal consequence; boundary consequence; direct materiality; indirect materiality; conditional materiality; cumulative materiality; systemic materiality; concentration materiality; critical assumptions; high-impact assumptions; low-probability high-consequence assumptions; threshold-based materiality; materiality uncertainty; materiality change; materiality reassessment; and historical materiality preservation.

Module Function

Not every assumption deserves the same level of governance attention.

A prediction may contain dozens or hundreds of assumptions.

Some may have almost no effect if they change.

Others may determine whether the prediction remains meaningful at all.

For example:

Assumption A: Average transport cost varies by $\pm 1\%$.

Assumption B: The only licensed facility remains authorized to operate.

Both assumptions may be explicit.

Both may be dependencies.

But failure of Assumption B may fundamentally invalidate the predicted operational pathway.

AMAM makes that difference governable.

Assumption Materiality

Assumption Materiality represents the significance of a Predictive Assumption to the legitimate meaning, structure, interpretation, use, or continuation of a prediction.

Materiality is determined by consequence.

It is not determined solely by:

assumption probability; assumption confidence; sensitivity magnitude; dependency strength; number of downstream relationships.

These may inform materiality but do not individually determine it.

Material Assumption

A Material Assumption is a Predictive Assumption whose change, weakening, breach, uncertainty, or invalidity could materially affect one or more governed properties of the prediction.

These may include:

predicted outcome; Prediction Formation conditions; methodology applicability; probability representation; uncertainty; confidence; feasibility; interpretation; reliance; continuing validity of the predictive structure.

Critical Assumption

A Critical Assumption is a Material Assumption whose failure or material change may fundamentally alter, invalidate, or severely restrict the prediction.

A Critical Assumption may affect:

whether the prediction can still be formed; whether the methodology remains applicable; whether the outcome remains feasible; whether stated probability remains meaningful; whether reliance should continue; whether prediction-level review becomes necessary.

Material Assumption $\neq$ Critical Assumption

Criticality represents a stronger materiality state.

Materiality Dimensions

Materiality may arise across several dimensions.

AMAM may evaluate:

outcome materiality; methodology materiality; probability materiality; uncertainty materiality; feasibility materiality; interpretation materiality; reliance materiality; temporal materiality; operational materiality;

systemic materiality.

An assumption may be low in one dimension and high in another.

Outcome Materiality

Outcome Materiality exists where assumption change may materially alter the predicted outcome.

Example:

A production forecast assumes that a major manufacturing line remains available.

Loss of that line may materially reduce expected output.

Methodology Materiality

An assumption may be material because the Prediction Methodology depends upon it.

Example:

A methodology assumes the underlying process remains sufficiently stationary.

If that assumption fails, the methodology may no longer be applicable even before the output visibly changes.

Stable Output $\neq$ Stable Methodological Legitimacy

Probability Materiality

An assumption may materially alter the probability associated with the prediction.

PPUS later governs probability representation.

AMAM determines whether the assumption's effect on that representation is materially significant.

Uncertainty Materiality

An assumption may materially increase uncertainty without changing the central prediction.

For example:

Expected Demand = 1,000 units

may remain unchanged while uncertainty expands dramatically.

A stable central estimate does not make the assumption immaterial.

Confidence Materiality

An assumption may materially affect confidence in the prediction even where predicted output remains similar.

Stable Prediction $\neq$ Stable Confidence

Feasibility Materiality

An assumption may be material because its failure makes the predicted outcome less realistically achievable or entirely infeasible.

PFES later establishes final feasibility state.

AMAM determines whether the assumption is materially relevant to that state.

Interpretation Materiality

Some assumptions determine how a prediction may legitimately be interpreted.

A prediction that originally represents:
expected future state
may become only:
conditional scenario
if a core assumption weakens.
That interpretive change can be material.

Reliance Materiality

An assumption may materially affect whether a prediction can be legitimately relied upon for:
investment; operational planning; healthcare; infrastructure; insurance; public policy; autonomous action;
other decisions.
PCRS later governs permitted reliance.
AMAM identifies material assumption relationships to reliance.

Direct Materiality

Direct Materiality exists where assumption change directly affects a governed property of the prediction.
Example:
Exchange rate remains below X
directly affects a margin forecast.

Indirect Materiality

Indirect Materiality operates through dependency chains.
An upstream assumption may appear remote but ultimately have significant downstream effect.
Indirect Materiality $\neq$ Low Materiality

Conditional Materiality

An assumption may become material only under certain conditions.
Example:
Energy-price assumption may be immaterial when production volume is low but highly material at full capacity.
Conditional Materiality $\neq$ Universal Materiality

Cumulative Materiality

Several individually modest assumptions may collectively create a material predictive effect.
For example:
slightly higher costs; slightly lower demand; slightly longer lead times; slightly lower productivity.
Individually, each may be low impact.
Together, they may materially alter the prediction.
Individually Immaterial $\neq$ Collectively Immaterial

Systemic Materiality

An assumption may become material because it affects many parts of the predictive structure.

Example:

Central infrastructure remains available

may support dozens of downstream assumptions.

Systemic position can increase materiality even when immediate sensitivity appears moderate.

Dependency Concentration and Materiality

A highly concentrated dependency may increase the significance of an assumption.

However:

Dependency Concentration $\neq$ Materiality by Default

AMAM must evaluate consequence, not merely structural centrality.

Sensitivity and Materiality

Sensitivity strongly informs materiality.

However:

High Sensitivity $\neq$ Automatic Materiality

and:

Low Sensitivity $\neq$ Automatic Immateriality

An assumption may be weakly sensitive numerically but highly material legally, operationally, methodologically, or in reliance terms.

Assumption Failure Probability

The likelihood that an assumption fails is separate from its materiality.

A highly likely assumption failure may have little consequence.

A very unlikely failure may have catastrophic predictive consequences.

Therefore:

Probability of Failure $\neq$ Materiality

Low-Probability High-Consequence Assumption

A Low-Probability High-Consequence Assumption is an assumption that is unlikely to fail but whose failure could materially or critically affect the prediction.

Examples may include:

failure of national infrastructure; loss of core legal authorization; collapse of a sole supplier; shutdown of a critical data service; catastrophic physical-system failure.

Such assumptions must not be treated as immaterial merely because failure appears unlikely.

Low Probability of Failure $\neq$ Low Assumption Materiality

High-Probability Low-Consequence Assumption

An assumption may be highly likely to change yet remain low-materiality where the predictive effect is limited.

Frequency does not equal importance.

Materiality Threshold

A Materiality Threshold establishes the point at which assumption consequence becomes sufficiently significant to require elevated governance treatment.

Thresholds may be:

quantitative; qualitative; operational; methodological; economic; probabilistic; temporal; reliability-based; consequence-based.

OOF® does not prescribe one universal threshold for all predictions.

Thresholds must fit predictive context.

Criticality Threshold

A Criticality Threshold represents a stronger boundary at which assumption failure may fundamentally change or invalidate the prediction.

For example:

Supplier capacity falls below 80%

may reduce output.

Below:

40%

the predicted production plan may become infeasible.

The latter may represent criticality.

Materiality State

A governed assumption may receive a Materiality State.

Potential states include:

Immaterial

Low Materiality

Moderate Materiality

High Materiality

Critical

Conditionally Material

Unable to Determine

The implementation may use another governed state model where appropriate.

Materiality Uncertainty

Materiality may itself be uncertain.

Possible reasons include:

insufficient dependency knowledge; uncertain sensitivity; unknown operating conditions; new methodology; incomplete Prediction Basis; limited historical information.

AMAM preserves uncertainty rather than forcing unjustified classification.

Materiality Assessment $\neq$ Materiality Certainty

Materiality Evidence Context

Materiality determinations may reference:

Prediction Basis; sensitivity assessments; operational conditions; methodology structure; historical outcomes; simulations; expert analysis.

AMAM governs the materiality determination.

It does not absorb ADIT® evidence governance or SIMULOS® simulation governance.

Materiality by Prediction Purpose

The same assumption may have different materiality depending upon Prediction Purpose.

Example:

A 2% demand deviation may be immaterial for:

long-term strategic planning

but material for:

same-day inventory allocation.

Same Assumption $\neq$ Same Materiality Across Purposes

Materiality by Prediction Horizon

Assumption materiality may change across Prediction Horizons.

A supplier-continuity assumption may have low materiality for a 24-hour forecast but high materiality for a 12-month production prediction.

Materiality at Horizon A $\neq$ Materiality at Horizon B

Materiality by Predictive Boundary

The same assumption may be material within one boundary and immaterial within another.

Example:

Weather may be highly material to agricultural output in one region but irrelevant to a separate digital-services forecast.

Materiality by Prediction State

Materiality may change as the prediction progresses through its lifecycle.

An assumption that was initially low-materiality may become critical after:

dependency change; system change; external event; reduction in alternatives; change in methodology; change in reliance context.

Materiality Interaction

Two assumptions may become material only in combination.

Example:

Supplier delay

alone may be manageable.

Supplier delay + low inventory

may become critical.

AMAM preserves materially relevant interaction effects.

Materiality Amplification

Dependency relationships may amplify assumption consequence.

A moderate upstream assumption may become highly material because many downstream assumptions rely upon it.

Materiality Dampening

Alternative pathways may reduce materiality.

For example:

A supplier-continuity assumption may be less material where three independently qualified backup suppliers exist.

Dependency $\neq$ Unavoidable Consequence

Alternative Pathways

Materiality should consider whether substitute pathways exist.

Possible alternatives include:

backup supplier; alternative infrastructure; alternate model; different operating mode; fallback process; alternative data source.

The existence of alternatives may reduce materiality but does not automatically eliminate it.

Assumption Concentration Risk

A prediction may be materially exposed where many critical conditions depend upon a small number of assumptions.

This does not itself determine final prediction risk.

AMAM identifies the material assumption structure.

Materiality and Prediction Feasibility

An assumption may be material because its failure reduces or eliminates the feasibility of the predicted outcome.

Example:

Required financing remains available.

If financing disappears, a project-completion prediction may become infeasible.

Materiality and Prediction Reliance

An assumption may be material because decisions are made on the prediction.

For example:

A prediction used only for exploratory research may tolerate greater assumption uncertainty than one used to authorize:

major investment; critical infrastructure action; autonomous execution.

Materiality can therefore depend upon reliance context.

Materiality and Predictive Reflexivity

Where predictions influence the environments they predict, assumptions about behavioral response may become especially material.

Example:

A market forecast may influence participant behavior.

An assumption of passive market response may therefore become material.

AMAM identifies the materiality relationship without taking over broader reflexivity governance.

Materiality Change

Assumption materiality may change when:

dependency structure changes; sensitivity changes; Prediction Purpose changes; Horizon changes; Predictive Boundary changes; methodology changes; Operational Reality changes; alternatives disappear; reliance increases.

Therefore:

Initial Materiality $\neq$ Continuing Materiality

Materiality Reassessment

Materiality should be reassessed where material changes occur in:

dependency structure; sensitivity profile; assumption state; Prediction Basis; methodology; Prediction Horizon; Predictive Boundary; operational context; reliance conditions.

Historical Materiality State

A historical prediction must preserve the materiality state applied at the relevant time.

A later determination that an assumption was actually more important must not silently rewrite the original governance record.

Current Materiality $\neq$ Historical Materiality

and:

Later Materiality Knowledge $\neq$ Original Materiality Determination

Materiality Governance Sufficiency

Assumption Materiality is sufficiently governed when:

materiality criteria are identifiable; materiality dimensions are understood; dependency and sensitivity context are considered; direct and indirect consequences are considered; low-probability high-consequence assumptions are not dismissed; cumulative and systemic effects are considered where relevant; thresholds are governed where applicable; critical assumptions are distinguishable; uncertainty remains visible; purpose, horizon, boundary, and reliance context are considered; materiality change can trigger reassessment; historical materiality states remain reconstructable.

Governance Requirements

AMAM requires that:

1. identified assumptions be assessed for materiality where relevant;
2. assumption presence not be treated as materiality by default;
3. materiality be determined by predictive consequence;
4. dependency strength remain distinguishable from materiality;
5. sensitivity magnitude remain distinguishable from materiality;
6. direct and indirect materiality be considered;
7. conditional materiality remain identifiable;
8. cumulative materiality be considered where relevant;
9. systemic materiality be considered where relevant;
10. probability of assumption failure remain separate from materiality;
11. low-probability high-

consequence assumptions not be dismissed; 12. materiality thresholds be defined where appropriate; 13. criticality remain distinguishable from ordinary materiality; 14. methodology, probability, uncertainty, feasibility, interpretation, and reliance consequences be considered where relevant; 15. alternative pathways be considered where material; 16. Prediction Purpose be considered in materiality determination; 17. Prediction Horizon be considered where material; 18. Predictive Boundary be considered where material; 19. materiality uncertainty remain explicit; 20. materiality interaction and amplification remain visible where relevant; 21. materiality change trigger reassessment where necessary; 22. current and historical materiality states remain distinguishable; 23. later knowledge not overwrite original materiality determinations.

Governance Logic

AMAM applies the following governance logic:

Identified Predictive Assumptions

Governed Assumption Dependency Map

Governed Assumption Sensitivity Profile

↓

Potential Predictive Consequences Identified

↓

Outcome / Methodology / Probability / Uncertainty / Feasibility / Interpretation / Reliance Consequences Assessed

↓

Direct / Indirect / Conditional Consequences Distinguished

↓

Cumulative / Systemic Effects Considered

↓

Failure Probability Distinguished from Consequence

↓

Materiality Threshold Applied where applicable

↓

Criticality Threshold Applied where applicable

↓

Materiality Uncertainty Preserved

↓

Materiality State Determined

↓

Critical Assumptions Identified

↓

Historical Materiality State Preserved

↓

Governed Assumption Materiality Profile

↓

ACGM — Assumption Change Governance Module

Where materiality cannot be sufficiently established:

Assumption Materiality Governance Gap → Additional Analysis / Qualification / Conditional Classification / Explicit Uncertainty / Restricted Interpretation

Minimum Implementation Framework

A minimum implementation of AMAM must establish:

A. Assumption Reference

The Predictive Assumption being assessed.

B. Dependency Reference

The applicable dependency context from ADMM.

C. Sensitivity Reference

The applicable sensitivity context from ASNM.

D. Materiality Dimensions

Where applicable:

Outcome / Methodology / Probability / Uncertainty / Confidence / Feasibility / Interpretation / Reliance / Operational / Systemic / Other

E. Consequence State

The expected consequence if the assumption materially changes or fails.

F. Materiality State

At minimum:

Immaterial / Low / Moderate / High / Critical / Conditionally Material / Unable to Determine

G. Failure Probability Context

Where available, represented separately from materiality.

H. Materiality Threshold

The threshold used where applicable.

I. Criticality Threshold

The point at which assumption consequence becomes critical where applicable.

J. Cumulative / Interaction Reference

Other assumptions materially contributing to combined consequence.

K. Alternative Pathway Reference

Any substitute pathway that may reduce consequence.

L. Materiality Uncertainty

Any unresolved uncertainty in the determination.

M. Reassessment Trigger

Conditions requiring materiality review.

N. Historical Materiality State

The materiality classification associated with the applicable prediction version and time.

Governance Outputs

AMAM may produce:

Governed Assumption Materiality Profile; Assumption Materiality Register; Material Assumption Register; Critical Assumption Register; materiality-dimension record; consequence profile; materiality-state determination; conditional-materiality record; low-probability high-consequence record; cumulative-materiality assessment; systemic-materiality assessment; Materiality Threshold Record; Criticality Threshold Record; alternative-pathway record; materiality-uncertainty record; materiality-reassessment trigger; historical materiality record; unresolved materiality requirements.

The primary governed output is:

Governed Assumption Materiality Profile

This becomes a principal input into:

ACGM — Assumption Change Governance Module.

Critical Distinctions

Assumption Presence $\neq$ Assumption Materiality

Material Assumption $\neq$ Critical Assumption

Dependency Strength $\neq$ Assumption Materiality

Dependency Concentration $\neq$ Materiality by Default

High Sensitivity $\neq$ Automatic Materiality

Low Sensitivity $\neq$ Automatic Immateriality

Probability of Failure $\neq$ Materiality

Low Probability of Failure $\neq$ Low Assumption Materiality

High Probability of Change $\neq$ High Materiality

Stable Output $\neq$ Stable Methodological Legitimacy

Stable Prediction $\neq$ Stable Confidence

Indirect Materiality $\neq$ Low Materiality

Conditional Materiality $\neq$ Universal Materiality

Individually Immaterial $\neq$ Collectively Immaterial

Same Assumption $\neq$ Same Materiality Across Purposes

Materiality at Horizon A $\neq$ Materiality at Horizon B

Dependency $\neq$ Unavoidable Consequence

Initial Materiality $\neq$ Continuing Materiality

Materiality Assessment $\neq$ Materiality Certainty

Current Materiality $\neq$ Historical Materiality

Later Materiality Knowledge $\neq$ Original Materiality Determination

Enterprise Application

AMAM enables organizations to determine which assumptions deserve the strongest governance attention.

Without materiality governance, organizations may:

monitor trivial assumptions intensely; ignore assumptions capable of invalidating a prediction; treat low-probability assumptions as unimportant; assume numerical sensitivity is the only form of consequence; fail to identify assumptions that determine methodology applicability or reliance.

AMAM creates a ranked and governable assumption structure for:

AI forecasting + strategic planning + financial prediction + insurance + industrial systems + supply chains + infrastructure + healthcare + markets + autonomous systems.

It enables downstream monitoring to focus on assumptions whose state can materially affect the prediction.

Use Case 1 — The Assumption Is Unlikely to Fail, but If It Does the Prediction Collapses

A logistics company predicts that regional deliveries will remain within target times for the next quarter.

One assumption states:

The only cross-border rail corridor used for 70% of critical shipments remains operational.

Historical data suggests that closure is highly unlikely.

The organization therefore initially treats the assumption as low risk.

However, sensitivity and dependency analysis show that a full corridor closure would remove the primary transport pathway and make the delivery prediction infeasible within the existing network.

The critical governance question is:

Can an assumption legitimately be treated as unimportant merely because its probability of failure is low when failure would fundamentally alter the predicted outcome?

AMAM classifies the assumption based on consequence rather than likelihood alone.

The assumption may therefore become:

Critical

even where failure probability remains low.

Low Probability of Failure $\neq$ Low Assumption Materiality

Use Case 2 — A Forecast Number Barely Changes, but the Assumption Determines Whether Anyone Should Rely on It

An AI system predicts a medical-service demand level of:

10,000 cases

for the coming period.

A material assumption states that the patient population represented in historical data remains sufficiently comparable to the population currently entering the system.

Sensitivity testing shows that moderate population change alters the central forecast by only 2%.

Numerically, the assumption appears low-sensitivity.

However, the population shift materially increases uncertainty and weakens the methodology's applicability to several subgroups.

The forecast may therefore no longer support the same operational reliance.

The critical governance question is:

Can an assumption be classified as immaterial because the headline prediction barely changes when its failure materially weakens methodology applicability and permitted reliance?

AMAM identifies the assumption as material even though direct numerical sensitivity is low.

Low Sensitivity $\neq$ Low Materiality by Default

Canonical Boundary

AMAM governs the determination, classification, dimensional assessment, thresholding, criticality, uncertainty, cumulative significance, systemic significance, reassessment, and historical preservation of Assumption Materiality. It does not govern initial assumption identification, dependency mapping, sensitivity assessment itself, actual Assumption Change Governance, Prediction Formation, probability representation, final feasibility determination, communication, reliance decisions, Operational Reality, or prediction-level revision.

AMAM establishes:

which assumptions matter enough that their state can materially alter the legitimate meaning or use of the prediction.

It does not determine:

what governance action must occur when those assumptions actually change.

Canonical Closing Statement

Not every assumption deserves equal governance attention, and probability of failure alone cannot determine importance.

AMAM identifies the assumptions whose condition can materially change what the prediction means, supports, or can legitimately be used for.

An assumption becomes material because of consequence — not because it happens to be visible.

It

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